Philipps-Universität Marburg
Fachbereich Mathematik & Informatik

Georg Wille
Volkmar Welker

Probabilistic Combinatorics SS24

Sheet 2

Due: 29. April at 19:00

(1) Let $n \geq 2$ and $\mathcal{A}$ a set of n-element subsets of $\mathbb{N}$ with $\#\mathcal{A} < 4^{n-1}$. Show: There is a map $f : \mathbb{N} \to [4]$ with $\#f(A) \geq 2$ for all $A \in \mathcal{A}$, i.e. a colouring of $\mathbb{N}$ with 4 colours, such that there is no monochromatic set in $\mathcal{A}$.

(2) Let $G = (V, E)$ be an undirected graph. Show via a probabilistic argument: There is a subset $E' \subseteq E$, such that $\#E' \geq \#E/2$ and the subgraph $G' := (V, E')$ is bipartite; i.e. there are $V_1, V_2 \subset V$ such that $V_1 \cap V_2 = \emptyset$, $V_1 \cup V_2 = V$ and $\#e \cap V_1 = \#e \cap V_2 = 1$ for all $e \in E'$.

(3) Let $\langle \cdot, \cdot \rangle$ be the standard dot product on $\mathbb{R}^n$. Show probabilistically: For vectors $v_1, ..., v_n \in \mathbb{R}^n$ with length $||v_i|| := \sqrt{\langle v_i, v_i \rangle} = 1$, $1 \leq i \leq n$, there are $\epsilon_1, ..., \epsilon_n \in \{\pm 1\}$ with

$$||\epsilon_1 v_1 + ... + \epsilon_n v_n|| \leq \sqrt{n}.$$

Hinweis:

- The usual proof relies on $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ for independant random variables X and Y.
- The corresponding statement with $\geq \sqrt{n}$ holds aswell.

(4) (Repetition of elementary probability theory) Let $(\Omega, \mathbb{P})$ be a discrete probability space and let $T, X_0, X_1, ...$ be real valued random variables on $(\Omega, \mathbb{P})$ such that:

- all expected values exist,
- the random variables are independant,
- the random variables $X_0, X_1, ...$ are identically distributed.
- $T(\omega) \in \mathbb{N}$ for all $\omega \in \Omega$.

Proof:

(i) Let T_n, $n \in \mathbb{N}$ be the indicator random variable of the event $\{\omega \in \Omega \mid T(\omega) = n\}$ and $S_n := X_1 + ... + X_n$. Then T_n and S_n are independant.

(ii) Define the random variable

$$\begin{aligned} S_T : \Omega &\to \mathbb{R} \\ \omega &\mapsto \sum_{i=0}^{T(\omega)} X_i(\omega). \end{aligned}$$

Show that $S_T = \sum_{n\in\mathbb{N}} S_n \cdot T_n$.

(iii) $\mathbb{E}[S_T]$ exists and $\mathbb{E}[S_T] = \mathbb{E}[T]\mathbb{E}[X_1]$.